Mobile Application Development – CA3: Review 1

Idea & AI Use-Case Proposal

Milestone: Problem definition, AI/ML use case, and project scope

Section	Details
Group Number	10
Team Members	1. Md Atiullah Ansari – 25MCAR0108 2. Aniwesh Aman – 25MCAR0304 3. Hardik Patel – 25MCAR0094
Title	AI-Powered Smart Parking Management and Parking Space Rental System
App Name	ParkIQ
Proposed AI/ML Feature	ParkIQ uses Google Gemini through Firebase AI Logic for two AI-driven functions. The AI Assistant understands natural-language parking queries and helps users check Bangalore parking availability using location, date and time. The parking-owner flow uses Gemini to analyze a real parking-area photo captured with CameraX and generate suggested parking listing details for review.
Why AI is Needed	AI makes parking search more interactive and reduces the effort needed to find parking. The AI Assistant can understand normal user questions and parking requests, while the owner feature reduces manual data entry by analyzing a real parking image and preparing a draft listing. The owner reviews and confirms AI-suggested values before saving.
Target Users	Drivers looking for parking in Bangalore; users who need to reserve parking; and parking owners who want to rent unused parking spaces.
Add-ons as per Remarks	Implemented the AI Assistant, Bangalore parking-availability search with Location/Date/Time options, real CameraX photo capture, Gemini image analysis for owner listings, Room Database persistence, retry and error handling, and the five-space parking-rental management flow.

Evaluation – Idea & AI Use-Case Proposal

Remarks:
The proposed AI features and project scope were clearly defined. The planned AI-based parking assistance and parking-owner support features were considered relevant and feasible for the ParkIQ application.

15/10/26

Signature of Members:

1. Md Atiullah Ansari MD. Atiullah Ansari
2. Aniwesh Aman Aniwesh Aman.
3. Hardik Patel

Signature of Faculty In charge:

Mr. Aldrin Jose Antony

Mobile Application Development – CA3: Review 2

AI/ML Model & Data Review

Milestone: Dataset preparation, model selection, and training approach

Section	Details
Group Number	10
Team Members	1. Md Atiullah Ansari – 25MCAR0108 2. Aniwesh Aman – 25MCAR0304 3. Hardik Patel – 25MCAR0094
Title	AI-Powered Smart Parking Management and Parking Space Rental System
App Name	ParkIQ
Dataset Used	A local ParkIQ parking dataset containing parking name, area/location, available slots, total slots, hourly price, rating, available hours and parking type. The application also stores owner-managed parking-space records in Room Database. The Bangalore AI search dataset covers locations including Indiranagar, BTM Layout, Jayanagar, Koramangala, HSR Layout, JP Nagar, Banashankari, Whitefield, Marathahalli, Bellandur, Electronic City, MG Road, Hebbal, Yelahanka, Rajajinagar, Malleshwaram, Vijayanagar, CV Raman Nagar, Frazer Town, Shivajinagar and Ulsoor.
Data Preprocessing	Parking records are organized by location and parking attributes. Common location-name variations are handled for searches. Owner records are persisted through Room Database. For image analysis, the actual CameraX-captured image is converted to image input and sent to Gemini.
Model / Algorithm Chosen	Google Gemini through Firebase AI Logic. A pre-trained multimodal Gemini model was chosen because it can process natural-language prompts and parking images without training a custom machine-learning model.
Training & Validation Approach	No custom model training is performed. Functional validation is used instead: normal chat queries, Bangalore location searches, Location/Date/Time filtering, no-data and non-Bangalore cases, and real parking-photo analysis are tested in the application.
Model Performance (so far)	Traditional accuracy, precision and recall are not directly applicable because a pre-trained generative AI service is used. Performance is evaluated through successful response generation, correct use of ParkIQ parking data, image-analysis behavior, and error/retry handling.
Integration Plan	Firebase AI Logic connects the Android app to Gemini. The AI Assistant sends user prompts to Gemini, while parking availability is read from ParkIQ data. The owner flow sends the real captured image to Gemini and displays the generated details for review before saving to Room Database.
Add-ons as per Remarks	Integrated Firebase Gemini, Room Database persistence, CameraX image capture, AI auto-fill for owner parking listings, persistent AI chat history, retry/error handling, and database-grounded parking availability.

Evaluation – AI Integration & Performance (partial)

Remarks:
The selected AI model and application-specific dataset were defined, and the AI integration approach was finalized. Data handling and validation methods were also planned for reliable AI-based parking assistance.

15/10/26

Signature of Members:

1. Md Atiullah Ansari Md. Atiullah Ansari
2. Aniwesh Aman Aniwesh Aman
3. Hardik Patel Hardik

Signature of Faculty In charge:

Mr. Aldrin Jose Antony

Mobile Application Development – CA3: Review 3

App Development & Functionality Review

Milestone: Core app build and functional integration of the AI feature

Section	Details
Group Number	10
Team Members	1. Md Atiullah Ansari – 25MCAR0108 2. Aniwesh Aman – 25MCAR0304 3. Hardik Patel – 25MCAR0094
Title	AI-Powered Smart Parking Management and Parking Space Rental System
App Name	ParkIQ
Features Implemented	Home; Explore; parking search; Google Maps/Places; Parking Details; slot/date/time/duration booking; payment; booking confirmation; My Bookings; Saved Parking; Profile; Settings; Parking Owner; five rental spaces; CameraX capture; Gemini photo analysis; AI auto-fill; Room Database; persistent AI Assistant chat; Location/Date/Time options; Bangalore parking availability; slot counts; no-data and non-Bangalore handling.
Tech Stack Used	Android Studio, Kotlin, Jetpack Compose, Material 3, Firebase AI Logic, Google Gemini, Room Database, CameraX, Google Maps SDK, Google Places API, Android Location Services.
AI Feature Functionality	AI Assistant: normal questions receive Gemini responses. Parking queries use Location, Date and Time options and ParkIQ data to show matching parking areas and available slots. Owner AI: capture a real parking-area photo with CameraX, send the actual image to Gemini, receive suggested listing details, review/edit and save.
Error Handling / Edge Cases	Loading and retry states; categorized AI errors; duplicate-request prevention; camera permission handling; empty AI fields before photo capture; no parking result; non-Bangalore result; persistent chat history; owner review before saving AI suggestions.
Known Issues / Bugs	Exact rating, rent and available slots cannot always be determined from a photo, so these values are AI-suggested/estimated and require owner review. Runtime AI behavior depends on Firebase/Gemini service availability.
Add-ons as per Remarks	Implemented the requested AI, database, camera, booking, owner-rental and error-handling enhancements. The project build was reported as successful with 0 compilation errors.

Evaluation – Application Functionality

Remarks:
The core ParkIQ application was implemented successfully with the major features integrated and functioning as planned. The AI features were connected with the application to support parking assistance and parking-space management.

Signature of Members:

1. Md Atiullah Ansari Md. Atiullah Ansari
2. Aniwesh Aman Aniwesh Aman.
3. Hardik Patel Hard

16/10/26

Signature of Faculty In charge:

Mr. Aldrin Jose Antony

Mobile Application Development – CA3: Review 4

UI/UX & AI Output Presentation Review

Milestone: Interface design and clarity of AI-generated results

Section	Details
Group Number	10
Team Members	1. Md Atiullah Ansari – 25MCAR0108 2. Aniwesh Aman – 25MCAR0304 3. Hardik Patel – 25MCAR0094
Title	AI-Powered Smart Parking Management and Parking Space Rental System
App Name	ParkIQ
Screens Designed	Home; Explore Parking; Parking Details; Parking Slot Selection; Date & Time; Confirm Booking; Payment; Booking Confirmed; My Bookings; Saved Parking; Profile; Settings; Parking Owner; My Parking Spaces; Add/Manage Parking Space; Real CameraX Capture; AI Photo Analysis/Review; AI Assistant.
Navigation Flow	Customer: Home or Explore → Parking Details → Select Slot → Date & Time → Duration → Confirm Booking → Payment → Booking Confirmed. AI Assistant: AI Assistant → Chat → Options → Location / Date / Time → Parking Availability Result. Owner: Home/Profile → Parking Owner → Select Parking Space → Real Camera → Capture Photo → Gemini AI Analysis → Review & Edit → Save.
AI Output Presentation	AI chat responses appear as chat bubbles. The Options control appears below the latest AI response and opens Location, Date and Time selectors. Parking availability is shown as parking names with available slot counts. Owner AI output is displayed in editable fields so the owner can review and confirm suggested values.
Interactive Components Used	Buttons; bottom navigation; search bars; parking cards; Options button; location-selection dialog; date picker; time picker; chat input; AI send button; loading indicator; retry button; CameraX preview; capture/shutter button; retake/use-photo controls; editable forms; Save button; dialogs; scrollable lists.
Responsiveness Check	Tested on Android device/emulator configurations including Pixel 8 and Motorola moto g96 5G, with attention to header positioning, drawable sizing, cards, navigation, scrolling and touch targets.
Add-ons as per Remarks	Implemented consistent header/drawable placement across Home, Explore and Profile, along with the requested AI Assistant, owner-management, booking and AI-output presentation adjustments.

Evaluation – UI/UX Design

Remarks:
The ParkIQ interface provides a clean and user-friendly design with consistent navigation and clearly presented AI-generated results. The AI features and interactive components are integrated to make parking search and management simple and easy to understand.

Signature of Members:

1. Md Atiullah Ansari Md. Atiullah Ansari
2. Aniwesh Aman Aniwesh Aman
3. Hardik Patel

Signature of Faculty In charge:

Mr. Aldrin Jose Antony

Mobile Application Development – CA3: Review 5

Final Deployment & Demo Review

Milestone: Deployed application, final AI performance, and live demo

Section	Details
Group Number	10
Team Members	1. Md Atiullah Ansari – 25MCAR0108 2. Aniwesh Aman – 25MCAR0304 3. Hardik Patel – 25MCAR0094
Title	AI-Powered Smart Parking Management and Parking Space Rental System
App Name	ParkIQ
Deployment Details	ParkIQ is developed as an Android application in Android Studio using Kotlin and Jetpack Compose. The current build is intended for academic demonstration and device/emulator testing; it is not deployed to the Google Play Store.
Final AI Model Performance	The application uses a pre-trained Gemini model through Firebase AI Logic, so a custom classification accuracy metric is not the primary measure. Final evaluation focuses on functional AI responses, database-grounded parking availability, real-image analysis, persistent chat history, and error/retry handling.
Final Feature Summary	Smart parking discovery; parking search; Google Maps/Places support; parking details; slot/date/time/duration booking; payment and booking confirmation; My Bookings; Saved Parking; Rewards; Profile and Settings; Parking Owner management; exactly five rental parking spaces; real CameraX photo capture; Gemini AI photo analysis and AI-suggested listing details; Room Database persistence; AI Assistant with persistent chat; Options for Location/Date/Time; Bangalore parking availability and available-slot results; no-data and non-Bangalore handling; retry and error handling.
Demo Notes	1) Show Home and the five rental parking spaces. 2) Demonstrate customer booking through Explore. 3) Open AI Assistant, type "Hi", show the natural AI response, open Options and select Location/Date/Time, then display parking availability and available slots. 4) Open Parking Owner, select a rental space, use the real camera to capture a parking-area photo, run Gemini analysis, review the generated details and save. 5) Show persisted owner data and persistent AI chat history.
Project Report Submitted	Yes – final project report to be attached/linked with the final CA3 submission. https://github.com/Atiullah18/ParkIQ
Add-ons as per Remarks	The requested AI Assistant, parking availability, owner AI image analysis, persistent storage, booking, and UI/UX features were incorporated into the final ParkIQ build for demonstration.

Final Evaluation – CA3 Rubric (Total 15 Marks)

15/10/26

Signature of Faculty In charge:

Mr. Aldrin Jose Antony

Signature of Members:

1. Md Atiullah Ansari MD. Atiullah Ansari
2. Aniwesh Aman Aniwesh Aman
3. Hardik Patel Hardik